FD-1 controller, Team Chakra, VK26-IITH-SM-0008. Sensor modules connect through headers J2-J7.
R8 = 33k sets the DRV8871 current limit near 2 A (check against datasheet before ordering).
J6 logic level: confirm MH-440D UART levels on bring-up; fit level shifter if 5 V.

+3V3 1
GND 2
SDA 3
SCL 4
J2
SDP810-500Pa dP

+3V3 1
GND 2
SDA 3
SCL 4
J3
SHT40 T/RH

+3V3 1
GND 2
SDA 3
SCL 4
J4
SEN0322 O2

+3V3 1
GND 2
SDA 3
SCL 4
J5
SSD1306 OLED

+5V 1
GND 2
NDIR_TX 3
NDIR_RX 4
NDIR_AN 5
J6
MH-440D CH4 NDIR

+5V 1
GND 2
P_RAW 3
J7
MPX5100AP pressure

+12V 1
GND 2
+5V 3
GND 4
U5
MP1584 buck module 12V>5V

VIN_RAW 1
GND 2
J15
12 V DC in

USB_DN 1
USB_DP 2
GND 3
J1
USB D-/D+ (native USB)

EN 1
BOOT 2
GND 3
J16
EN / BOOT / GND

+3V3 1
GND 2
SD_CS 3
SD_SCK 4
SD_MOSI 5
SD_MISO 6
J14
microSD module

485_RO 1
485_DE 2
485_DE 3
485_DI 4
485_A 1
485_B 2
GND 3
J8
RS-485 to gateway

+12V 1
SOL_DRV 2
J9
NC solenoid 12 V

SERVO_PWM 1
V_SERVO 2
GND 3
J10
Throttle servo

V_SERVO 1
GND 2
J11
Servo 6 V BEC in

ROUT1 1
ROUT2 2
J12
3-way router valve

LR_O 1
LA_O 2
LG_O 3
BZ_O 4
+12V 5
J13
Stack light + buzzer

R1 4.7k
R2 4.7k
R3 10k
R4 10k
R5 120R
R6 100R
R7 100k
R8 33k
R9 10k
R10 10k

C1 100nF
C2 100nF
C3 1uF
C4 470uF 25V
C5 10uF
C6 22uF
C7 100nF

D1 1N5822
D2 SS34
D3 SMBJ15A
F1 2A

Q1 IRLZ44N

U1 ESP32-S3-WROOM-1-N16R8
U2 MAX485
U3 DRV8871
U4 ULN2003A
U6 AMS1117-3.3

Vishwakarma Awards 2026-27, VK26-IITH-SM-0008
Team Chakra / Firedamp Dynamics
Sheet: /
File: fd1_controller.kicad_sch
Title: FD-1 Borehole Sentinel - node controller
Size: A3 | Date: 2026-10-22 | **Rev: B**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 1/1